Linear models for regression

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Supervised learning: regression

Machine learning by **induction from labeled data**

$$\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^{N}$$

$\mathbf{x} \in \mathcal{X}$: vector of **attributes**, features, independent variables, input variables. covariates…

$\mathbf{x}_n^T = (x_{n1} \;\; x_{n2} \;\; \dots \; x_{nD})$ [D-dimensional vector]

$y \in \mathcal{Y}$: (class) **label**, dependent variable, outcome, target, …

Regression: **Labels** y are **continuous** $(e.g. y \in \mathbb{R})$

Learning algorithm: $\mathcal{L}$

$\boldsymbol{\theta}$: Model parameters

$\mathcal{L}$: $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^{N} \rightarrow h(\,\cdot\,;\boldsymbol{\theta})$

$h(\,\cdot\,;\boldsymbol{\theta})$: $\mathbf{x} \in \mathcal{X} \;\rightarrow\; h(\mathbf{x};\boldsymbol{\theta}) \in \mathcal{Y}$

Linear models for regression

$\mathbf{w}^{\mathrm{T}} = (w_1, w_2, \dots, w_D)$
$\mathbf{x}^{\mathrm{T}} = (x_1\ x_2 \dots x_D)$

Learning to build a model from labeled data

$\mathcal{L}$: $\mathcal{D} = \{(\mathbf{x}_n, y_n)\}_{n=1}^{N} \rightarrow\ h(\mathbf{x}; w_0, \mathbf{w}) = w_0 + \mathbf{w}^T\mathbf{x}$

w_0: Independent term $\qquad$ $\mathbf{w}$: Linear coefficients

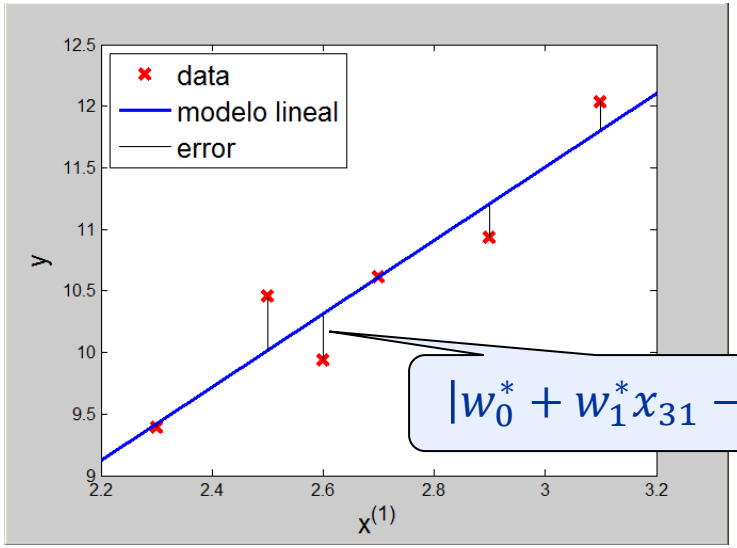
- $\mathcal{L}$: Learning algorithm
- **Loss (error) function**: $L(w_0, \mathbf{w}) = \mathbb{E}[L(h(\mathbf{X}\,; w_0, \mathbf{w}), Y)]$
- **Complexity penalty** (**regularization**) terms: $\{R_i(\mathbf{w})\}_{i=1}^{I}$
- **Learning = Training = Finding the model parameters**

$$(w_0^*, \mathbf{w}^*) = \underset{(w_0, \mathbf{w})}{\arg\min}\left(L(w_0, \mathbf{w}) + \sum_{i=1}^{I} \lambda_i\, R_i(\mathbf{w}) \right)$$

- Hyperparameters $\{\lambda_i\}_{i=1}^{I}$: weights of the regularization terms

Simple regression example

n	x_n	y_n
1	2.3	9.3915
2	2.5	10.4581
3	2.6	9.9371
4	2.7	10.6172
5	2.9	10.9357
6	3.1	12.0390



$$|w_0^* + w_1^* x_{31} - y_3|$$

Simple regression

Linear predictor (regressor)

$$h(x; w_0, w_1) = w_0 + w_1 x$$

w_0: Intercept $\qquad$ w_1: Slope

- **Labeled data**: $\mathcal{D} = \{(x_n, y_n)\}_{n=1}^{N}$
- **Model prediction:** $h(x_n; w_0, w_1) = w_0 + w_1 x$
- **Residual:** $e_n = y_n - h(x_n; w_0, w_1) = y_n - (w_0 + w_1 x_n)$
- **Loss functions**:

$$MSE(w_0, w_1) = \mathbb{E}\left[\left(Y - (w_0 + w_1 X)\right)^2\right] \approx \frac{1}{N}\sum_{n=1}^{N}\left(y_n - (w_0 + w_1 x_n)\right)^2$$

$$MAE(w_0, w_1) = \mathbb{E}[|Y - (w_0 + w_1 X)|] \approx \frac{1}{N}\sum_{n=1}^{N}|y_n - (w_0 + w_1 x_n)|$$

Simple regression: Least squares

Linear predictor (regressor)

$$h(x; w_0, w_1) = w_0 + w_1 x$$

- **Loss function:** $MSE(w_0, w_1) = \frac{1}{N}\sum_{n=1}^{N}\left((w_0 + w_1 x_n) - y_n\right)^2$
- **Minimization of empirical loss function**

$$\frac{\partial}{\partial w_0} MSE(w_0, w_1)\Big|_{\substack{w_0=w_0^* \\ w_1=w_1^*}} = 0 \quad \Rightarrow \quad \sum_{n=1}^{N}(w_0^* + w_1^* x_n - y_n) = 0 \quad \Rightarrow w_0^* = \bar{y} - w_1^* \bar{x}$$

$$\bar{x} = \frac{1}{N}\sum_{n=1}^{N} x_n\,; \qquad \bar{y} = \frac{1}{N}\sum_{n=1}^{N} y_n$$

$$\frac{\partial}{\partial w_1} MSE(w_0, w_1)\Big|_{\substack{w_0=w_0^* \\ w_1=w_1^*}} = 0 \Rightarrow \sum_{n=1}^{N} x_n(w_0^* + w_1^* x_n - y_n) = 0 \Rightarrow w_1^* = \frac{\frac{1}{N}\sum_{n=1}^{N}\hat{x}_n\hat{y}_n}{\frac{1}{N}\sum_{n=1}^{N}\hat{x}_n\hat{x}_n}$$

$$\hat{x}_n = x_n - \bar{x}; \qquad \hat{y}_n = y_n - \bar{y}$$

Multivariate linear regression (D dimensions)

Linear predictor (regressor)

$$h(\mathbf{x}; w_0, \boldsymbol{w}) = w_0 + \mathbf{w}^T\mathbf{x} = w_0 + (w_1 \dots w_D)\begin{pmatrix} x_1 \\ \vdots \\ x_D \end{pmatrix} = w_0 + w_1 x_1 + \cdots + w_D x_D$$

- **Empirical loss function**: $MSE(w_0, \mathbf{w}) = \frac{1}{N}\sum_{n=1}^{N}\left((w_0 + \mathbf{w}^T\mathbf{x}_n) - y_n\right)^2$
- **Minimization of empirical loss function:** Strictly convex (unique solution)

$$\frac{\partial}{\partial w_0} MSE(w_0, \mathbf{w})\Big|_{\substack{w_0 = w_0^* \\ \boldsymbol{w} = \boldsymbol{w}^*}} = 0 \quad \Rightarrow \quad \sum_{n=1}^{N}\left((w_0^* + (\mathbf{w}^*)^T\mathbf{x}_n) - y_n\right) = 0$$

$$\boxed{w_0^* = \bar{y} - (\mathbf{w}^*)^T\bar{\boldsymbol{x}}} \qquad \bar{\boldsymbol{x}} = \frac{1}{N}\sum_{n=1}^{N}\mathbf{x}_n\,; \qquad \bar{y} = \frac{1}{N}\sum_{n=1}^{N} y_n$$

- **Define the zero mean quantities**

$$\hat{\boldsymbol{x}}_n = \boldsymbol{x}_n - \bar{\boldsymbol{x}}; \qquad \hat{y}_n = y_n - \bar{y}$$

Multivariate linear regression (D dimensions)

$$MSE(w_0, \boldsymbol{w}) = \frac{1}{N}\sum_{n=1}^{N}\left((w_0 + \mathbf{w}^T\mathbf{x}_n) - y_n\right)^2$$

- **Minimization of empirical loss function**

$$(\mathbf{w}^*)^T\mathbf{x}_n = \mathbf{x}_n^T\mathbf{w}^*$$

$$\frac{\partial}{\partial \boldsymbol{w}^T} MSE(w_0, \mathbf{w})\Big|_{\substack{w_0 = w_0^* \\ \mathbf{w} = \mathbf{w}^*}} = 0 \quad \Rightarrow \quad \sum_{n=1}^{N}\mathbf{x}_n\left((w_0^* + (\mathbf{w}^*)^T\mathbf{x}_n) - y_n\right) = 0$$

$$\Rightarrow \quad \sum_{n=1}^{N}\hat{\mathbf{x}}_n(\hat{\mathbf{x}}_n^T\mathbf{w}^* - \hat{y}_n) = 0$$

Zero-mean quantities
$\hat{\boldsymbol{x}}_n = \boldsymbol{x}_n - \bar{\boldsymbol{x}}$;
$\hat{y}_n = y_n - \bar{y}$

$$\hat{\mathbf{X}}^T = (\hat{\mathbf{x}}_1 \dots \hat{\mathbf{x}}_N)$$

$$\hat{\mathbf{y}} = \begin{pmatrix} \hat{y}_1 \\ \vdots \\ \hat{y}_N \end{pmatrix}$$

$$\boxed{\mathbf{w}^* = \left(\sum_{n=1}^{N}\hat{\mathbf{x}}_n\hat{\mathbf{x}}_n^T\right)^{-1}\left(\sum_{n=1}^{N}\hat{\mathbf{x}}_n\hat{y}_n\right) = (\hat{\mathbf{X}}^T\hat{\mathbf{X}})^{-1}\hat{\mathbf{X}}^T\,\hat{\mathbf{y}}}$$

Multivariate regression: linear equations

Linear predictor of $\hat{y} = y - \bar{y}$

$$\mathbf{w}^T = (w_1 \dots w_D)$$
$$\hat{\mathbf{x}}^T = (\hat{x}_1 \dots \hat{x}_D)$$

$$\hat{h}(\hat{\mathbf{x}};\ \boldsymbol{w}) = \mathbf{w}^T \hat{\mathbf{x}} = w_1 \hat{x}_1 + \cdots + w_D \hat{x}_D$$

Design matrix (aka *data matrix, regressor matrix*)

$$\begin{aligned} w_1\hat{x}_{11} + w_2\hat{x}_{12} + \cdots + w_D\hat{x}_{1D} &= \hat{y}_1 \\ w_1\hat{x}_{21} + w_2\hat{x}_{22} + \cdots + w_D\hat{x}_{2D} &= \hat{y}_2 \\ \cdots \\ w_1\hat{x}_{N1} + w_2\hat{x}_{N2} + \cdots + w_D\hat{x}_{ND} &= \hat{y}_N \end{aligned} \Rightarrow \begin{pmatrix} \hat{x}_{11} & \hat{x}_{12} & \cdots & \hat{x}_{1D} \\ \hat{x}_{21} & \hat{x}_{22} & \cdots & \hat{x}_{2D} \\ \vdots & & \ddots & \vdots \\ \hat{x}_{N1} & \hat{x}_{N2} & \cdots & \hat{x}_{ND} \end{pmatrix} \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_D \end{pmatrix} = \begin{pmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_N \end{pmatrix}$$

$\hat{\mathbf{X}}\,\mathbf{w} = \hat{\mathbf{y}}$ **Typically,** $N \gg D$ The system is overdetermined

(more equations that unknowns)

$$\hat{\mathbf{X}}^T \hat{\mathbf{X}}\,\mathbf{w}^* = \hat{\mathbf{X}}^T \hat{\mathbf{y}} \qquad \Rightarrow \qquad \boldsymbol{w}^* = \left(\hat{\mathbf{X}}^T \hat{\mathbf{X}}\right)^{-1} \hat{\mathbf{X}}^T \hat{\mathbf{y}}$$

Compute me using singular value decomposition (SVD)

$\left(\hat{\mathbf{X}}^T \hat{\mathbf{X}}\right)^{-1} \hat{\mathbf{X}}^T$ **: Moore-Penrose pseudo-inverse**

Multivariate linear regression with bias term

$\mathbf{w}^T = (w_0, w_1 \dots, w_D)$
$\mathbf{x}^T = (x_0, x_1 \dots, x_D)$

$x_0 = 1$ (bias term)

Linear predictor of y

$$h(\mathbf{x}; w_0, \boldsymbol{w}) = \mathbf{w}^T\mathbf{x} = w_0 x_0 + w_1 x_1 + \cdots + w_D x_D$$

$$\begin{aligned} w_0 + w_1 x_{11} + w_2 x_{12} + \cdots + w_D x_{1D} &= y_1 \\ w_0 + w_1 x_{21} + w_2 x_{22} + \cdots + w_D x_{2D} &= y_2 \\ \cdots \\ w_0 + w_1 x_{N1} + w_2 x_{N2} + \cdots + w_D x_{ND} &= y_N \end{aligned} \Rightarrow \begin{pmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1D} \\ 1 & x_{21} & x_{22} & \cdots & x_{2D} \\ & \vdots & & \ddots & \vdots \\ 1 & x_{N1} & x_{N2} & \cdots & x_{ND} \end{pmatrix} \begin{pmatrix} w_0 \\ w_1 \\ \vdots \\ w_D \end{pmatrix} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_N \end{pmatrix}$$

$\mathbf{X}\,\mathbf{w} = \boldsymbol{y}$ **Typically,** $N \gg D$ The system is overdetermined

(more equations that unknowns)

$$\mathbf{X}^T\mathbf{X}\,\mathbf{w}^* = \mathbf{X}^T\mathbf{y} \qquad \Rightarrow \qquad \mathbf{w}^* = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$$

$\mathbf{w}^*[1{:}D]$ is the same solution as in previous slide.

Regularized Least Squares

Regularization: Include in the cost function terms that penalize the complexity of the model

Why is regularization necessary?

- There are **more variables than observations** ($D \gg N$)
  LS estimator may not be unique (e.g. microarray data)
- **Numerical instabilities** of least squares:
  $\left(\widehat{\mathbf{X}}^T \widehat{\mathbf{X}}\right)$ can be close to singular, in which case, small changes in $\widehat{\mathbf{X}}$ can lead to large changes in $\left(\widehat{\mathbf{X}}^T \widehat{\mathbf{X}}\right)^{-1}$
- **Overfitting avoidance**: By penalizing the complexity of the model it is possible to obtain more robust models that generalize well.
- **Parsimony & interpretability**: Some regularization methods (namely, lasso, elastic net) can lead to models in which some regression coefficients are exactly zero. These a simpler models can bring more light on the relationship between the response and the inputs.

L_2 regularization: Ridge regression

IMPORTANT: Centering and scaling of attributes $\bar{\mathbf{x}} = \mathbf{0}$; $std(\mathbf{x}) = \mathbf{1}$
Assuming $\bar{y} = 0$, the independent term is $w_0^* = \bar{y} - (\boldsymbol{w}^*)^T \bar{\boldsymbol{x}} = 0$

Linear predictor (regressor)
$h(\mathbf{x}; \boldsymbol{w}) = \boldsymbol{w}^T \mathbf{x} = w_1 x_1 + \cdots + w_D x_D$

- **Sum of squared errors**: $E(\boldsymbol{w}) = \sum_{n=1}^{N} (\boldsymbol{w}^T \mathbf{x}_n - y_n)^2$
- **L_2 penalty:** $R_{L_2}(\boldsymbol{w}) = w_1^2 + w_2^2 + \cdots + w_D^2 = \sum_{d=1}^{D} w_d^2 = \boldsymbol{w}^T \boldsymbol{w} = \|\boldsymbol{w}\|_2^2$
- **Cost function:** Strictly convex (unique solution)

To be determined e.g. by CV

$$C(\boldsymbol{w}) = E(\boldsymbol{w}) + \lambda_2 R_{L_2}(\boldsymbol{w}) = \sum_{n=1}^{N} (\boldsymbol{w}^T \mathbf{x}_n - y_n)^2 + \lambda_2 \boldsymbol{w}^T \boldsymbol{w}$$

- **Minimization of cost function**

$$\frac{\partial}{\partial \boldsymbol{w}^T} C(\boldsymbol{w}) \Big|_{\boldsymbol{w}=\widehat{\boldsymbol{w}}} = 0 \Rightarrow \sum_{n=1}^{N} \mathbf{x}_n (\mathbf{x}_n^T \boldsymbol{w}^* - y_n) + \lambda_2 \boldsymbol{w}^* = 0$$

$D \times D$ identity matrix

$$\boldsymbol{w}^* = \left(\sum_{n=1}^{N} \mathbf{x}_n \mathbf{x}_n^T + \lambda_2 \mathbf{I} \right)^{-1} \left(\sum_{n=1}^{N} \mathbf{x}_n y_n \right) = (\mathbf{X}^T \mathbf{X} + \lambda_2 \mathbf{I})^{-1} \mathbf{X}^T \mathbf{y}$$

Ridge regression: bias + variance

$$MSE = Bias^2 + \text{Variance}$$

Image from (Hoerl & Kennard, 1970)

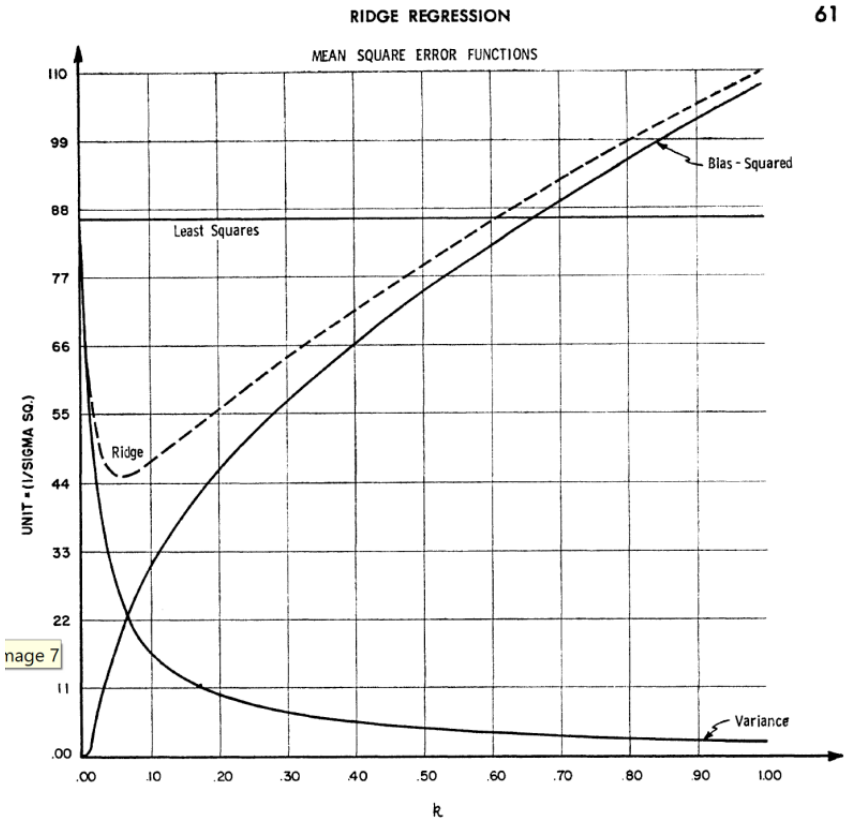


FIGURE 1

L_1 regularization: Lasso

Linear predictor (regressor): $h(\mathbf{x}; \boldsymbol{w}) = \boldsymbol{w}^T\mathbf{x} = w_1x_1 + \cdots + w_Dx_D$

- **Sum of squared errors**: $E(\boldsymbol{w}) = \sum_{n=1}^{N}(\boldsymbol{w}^T\mathbf{x}_n - y_n)^2$
- **L_1 penalty:** $R_{L_1}(\boldsymbol{w}) = |w_1| + |w_2| + \cdots + |w_D| = \sum_{d=1}^{D}|w_d| = \|\boldsymbol{w}\|_1$
- **Cost function:** Strictly convex (unique solution)

$$C(\boldsymbol{w}) = E(\boldsymbol{w}) + \lambda_1 R_{L_1}(\boldsymbol{w}) = \sum_{n=1}^{N}(\boldsymbol{w}^T\mathbf{x}_n - y_n)^2 + \lambda_1\|\boldsymbol{w}\|_1$$

To be determined e.g. by cross validation

- **Minimization of cost function**

$$\boldsymbol{w}^* = \arg\min_{\boldsymbol{w}}\left(\sum_{n=1}^{N}(\boldsymbol{w}^T\mathbf{x}_n - y_n)^2 + \lambda_1\|\boldsymbol{w}\|_1\right)$$

Feature selection: For sufficiently large values of λ_1 some components of $\boldsymbol{w}^*$ are exactly zero.

- Equivalent to linear SVM: Use of fast SVM algorithms for Lasso regression (Jaggi, 2014)

Lasso: regularization path

Equivalent formulation of Lasso

$$\mathbf{w}^* = \arg\min_{\mathbf{w}} \sum_{n=1}^{N} (\mathbf{w}^T \mathbf{x}_n - y_n)^2$$

$$\text{s.t.} \quad \sum_{d=1}^{D} |w_d| \leq t$$

$$s = \frac{t}{\sum_{i=1}^{D} |w_i^{LS}|}$$

$\mathbf{w}^{LS}$: Vector of least-squares weights

Figure from Tibshirani, (1996)

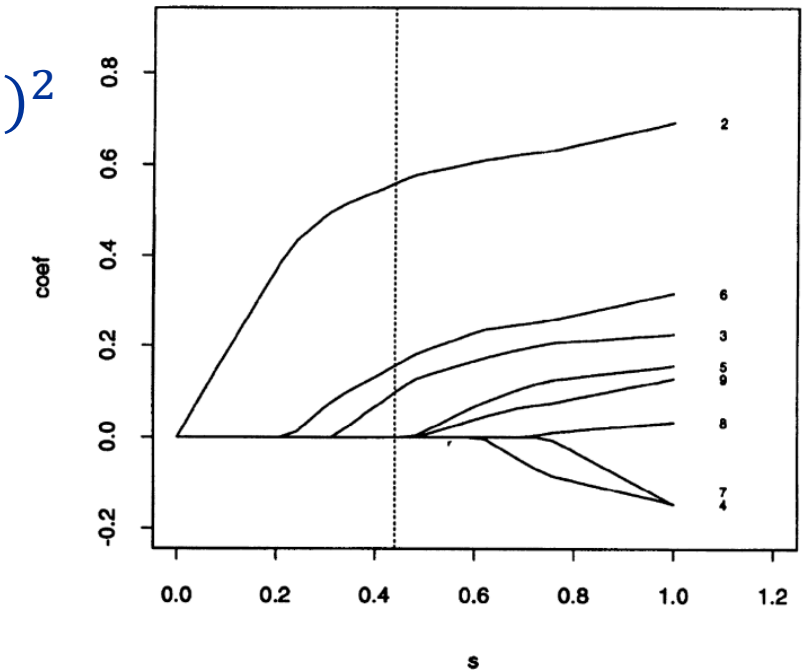


Fig. 5. Lasso shrinkage of coefficients in the prostate cancer example: each curve represents a coefficient (labelled on the right) as a function of the (scaled) lasso parameter $s = t/\Sigma|\beta_j^o|$ (the intercept is not plotted); the broken line represents the model for $\hat{s} = 0.44$, selected by generalized cross-validation

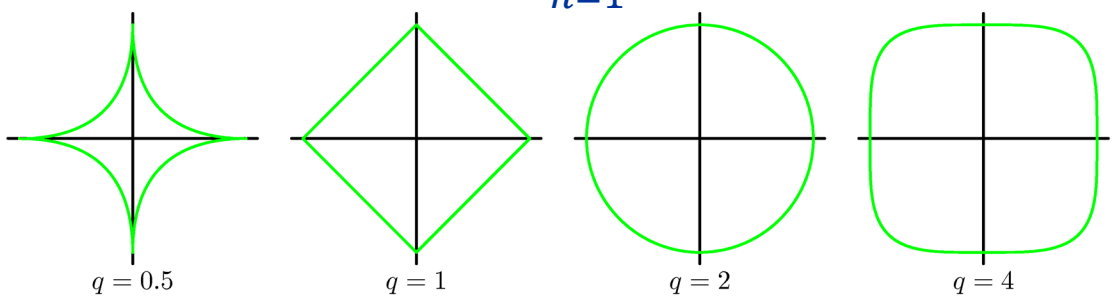
L_q regularization

$$\|\boldsymbol{w}\|_p^p = \sum_{d=1}^{D} w_d^p$$

- **Cost function = Error + L_q penalty term**

$$C(\mathbf{w}) = E(\mathbf{w}) + \lambda_p R_{L_q}(\mathbf{w}) = \sum_{n=1}^{N} (\mathbf{w}^T \mathbf{x}_n - y_n)^2 + \lambda_q \|\boldsymbol{w}\|_q^q$$

Bishop PRML 2006
Fig. 3.3



- **Geometrical interpretation**

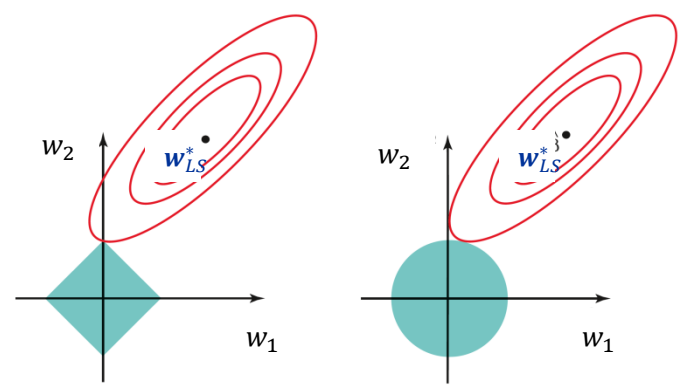


FIGURE 6.7. *Contours of the error and constraint functions for the lasso* (left) *and ridge regression* (right). *The solid blue areas are the constraint regions,* $|\beta_1| + |\beta_2| \le s$ *and* $\beta_1^2 + \beta_2^2 \le s$, *while the red ellipses are the contours of the RSS.*

Image from "An Introduction to Statistical Learning, with Applications in R"
R. G. James, D. Witten, T. Hastie, and R. Tibshirani

Lasso shortcomings

From (Zou & Hastie, 2005):

(a) In the $D > N$ case, the lasso **selects at most** N **variables** before it saturates, because of the nature of the convex optimization problem. This seems to be a limiting feature for a variable selection method. Moreover, the lasso is not well defined unless the bound on the L1-norm of the coefficients is smaller than a certain value.

(b) If there is a group of variables among which the pairwise correlations are very high, then the **lasso tends to select only one variable from the group** and does not care which one is selected. See Section 2.3.

(c) For usual $N > D$ situations, if there are **high correlations between predictors**, it has been empirically observed that the prediction performance of the **lasso is dominated by ridge regression** (Tibshirani, 1996).

$L_1 + L_2$ regularization: elastic net

Alternative hyperparameters:
$\lambda_1 = \alpha\,\lambda; \qquad \lambda_2 = (1-\alpha)\,\lambda$

- **Cost function:**

$$C(\boldsymbol{w}) = E(\boldsymbol{w}) + \lambda_1 R_{L_1}(\boldsymbol{w}) + \lambda_2 R_{L_2}(\boldsymbol{w})$$
$$= \sum_{n=1}^{N} (\boldsymbol{w}^T \mathbf{x}_n - y_n)^2 + \lambda_1 \|\boldsymbol{w}\|_1 + \lambda_2 \|\boldsymbol{w}\|_2^2$$

Strictly convex
(unique solution)

λ_1, λ_2 to be determined e.g. by CV

- **Minimization of cost function**

$$\boldsymbol{w}^* = \arg\min_{\boldsymbol{w}} \left(\frac{1}{2N} \sum_{n=1}^{N} (\boldsymbol{w}^T \mathbf{x}_n - y_n)^2 + \lambda_1 \|\boldsymbol{w}\|_1 + \lambda_2 \|\boldsymbol{w}\|_2^2 \right)$$

Feature selection: For sufficiently large values of λ_1 some components of $\boldsymbol{w}^*$ are exactly zero.

- Equivalent to linear SVM: Use of fast SVM algorithms for Lasso regression (Zhou et al, 2015)

Generalized Lasso

$$C(\mathbf{w}) = \sum_{n=1}^{N} (\mathbf{w}^T \mathbf{x}_n - y_n)^2 + \lambda_1 \|\mathbf{D}\mathbf{w}\|_1$$

D: general penalty matrix

- Standard Lasso: $\mathbf{D} = \mathbf{I}$
- 1D fused Lasso

Smooth variation of coefficients

$$\mathbf{D} = \begin{pmatrix} -1 & 1 & 0 & \cdots & 0 & 0 \\ 0 & -1 & 1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 1 \end{pmatrix}$$

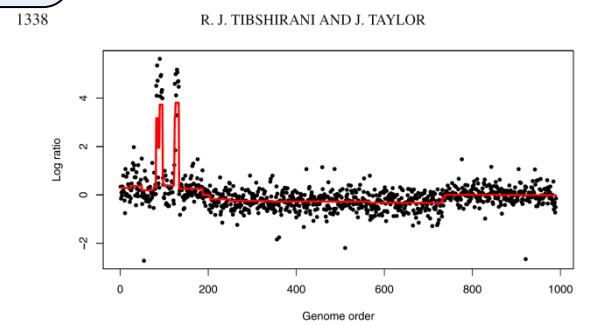


FIG. 1. *The 1d fused lasso applied to some glioblastoma multiforme data. The red line represents the inferred copy number from the 1d fused lasso solution (for $\lambda = 3$).*

- 2D fused Laso for images:

  D is such that differences between horizontal /vertical pixels are penalized (*total variation denoising*)

  Efficient algorithm (Tibshirani & Taylor 2011)

Group Lasso

G groups of coefficients: $\mathbf{w}^{\mathrm{T}} = \left(\left(\mathbf{w}^{(1)}\right)^{\mathrm{T}}\left(\mathbf{w}^{(2)}\right)^{\mathrm{T}} \dots \left(\mathbf{w}^{(G)}\right)^{\mathrm{T}}\right)$

$$\left(\mathbf{w}^{(g)}\right)^{\mathrm{T}} = \left(\left(w_1^{(g)}\right)^{\mathrm{T}}\left(w_2^{(g)}\right)^{\mathrm{T}} \dots \left(w_{d_g}^{(g)}\right)^{\mathrm{T}}\right); \qquad g = 1,2,\dots,G$$

$$d_1 + d_2 + \cdots + d_G = D$$

- Group lasso (Yuan & Lin, 2006)

$$C(\mathbf{w}) = \frac{1}{2}\sum_{n=1}^{N}(\mathbf{w}^T\mathbf{x}_n - y_n)^2 + \lambda_1 \sum_{g=1}^{G}\sqrt{d_g}\,\|\mathbf{w}^{(g)}\|_2$$

Either all zero or all non-zero within group

$$\|\mathbf{w}^{(g)}\|_2 = \sqrt{\left(w_1^{(g)}\right)^2 + \left(w_2^{(g)}\right)^2 + \cdots + \left(w_{d_g}^{(g)}\right)^2}$$

Discontinuous derivative when $\|\mathbf{w}^{(g)}\|_2 = 0$

Discontinuous derivative if $w_i^{(g)} = 0$

- Sparse group lasso (Simon et al, 2012): Some $w_i^{(g)}$ within g can be zero

$$C(\mathbf{w}) = \frac{1}{2N}\sum_{n=1}^{N}(\mathbf{w}^T\mathbf{x}_n - y_n)^2 + (1-\alpha)\lambda_1 \sum_{g=1}^{G}\sqrt{d_g}\,\|\mathbf{w}^{(g)}\|_2 + \alpha\lambda_1\|\mathbf{w}\|_1$$

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Multivariate regression issues

- **Robustness to outliers (MSE vs MAE)**
- **Uncertainty intervals for parameters**
- **T-test for significance of parameters**
- **Approximation vs interpolation**
- **Polynomial interpolation bias vs variance**